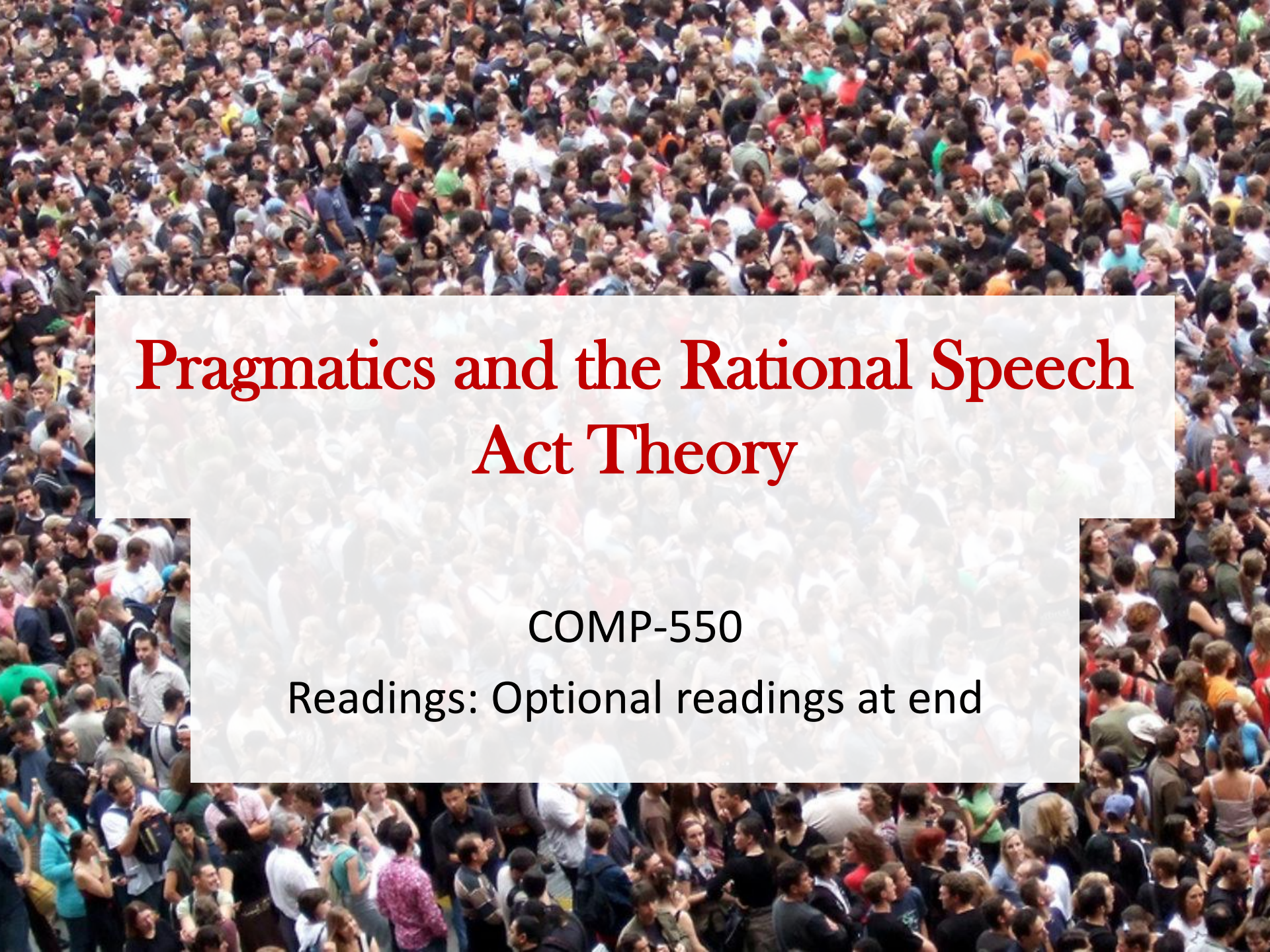




Pragmatics and the Rational Speech Act Theory

COMP-550

Readings: Optional readings at end

Outline

Pragmatics

- Layers of meaning

- Implicatures, performatives

Speech acts

- From truth to felicity

- The King of France returns

Rational speech act theory

- Grice and the cooperative principle

- Probabilistic pragmatics

- Scalar implicature

Semantics – Literal Meaning

Primary concern is truth conditions in the world

Flashback

Relates the linguistic expression to the world:

- Asserts a proposition that is either true or false relative to the world

Pandas are purple and yellow.

- Conveys information about the world

It will snow tomorrow.

- Is a query about the state of the world

What is the weather like in Montreal next week?

What is the Meaning of the Following?

Can you pass me the salt?

I am sorry.

I declare this store to be open.

I am pleased to introduce the next speaker.

Whether the statements are literally true or false is not the most relevant or important aspect of the meaning!

Implicature

Implying a meaning that is different from what is literally said.

Can you pass me the salt?

→ *I politely request you to pass me the salt.*

implicates

Can be cancelled (unlike entailment)!

Can you pass me the salt? As in, do you have the strength to do so? I don't actually want it.

Performative Utterances

When saying something is itself an act of doing something (Austin, 1962).

I am sorry. : an act of apologizing

I am pleased to introduce the next speaker.

: an act of introduction

Beyond Truth

For sentences like the performatives above, it is difficult to define whether something is “true” or not.

- *I am sorry.*

True as in I apologized? Or as in an internal feeling?

Instead of discussing **true** vs. **false**, we need another notion of whether the utterance is appropriate in its context: **felicity**

Felicity Conditions (Inexhaustive)

Existence condition – an accepted conventional procedure having a certain effect exists

- E.g., apologizing is accepted as a form of redress for a wrong

Appropriateness condition – context matches what is expected for convention to trigger

- E.g., speaker did something wrong or regrettable

Sincerity condition – thoughts and intentions matter!

- E.g., speaker has an internal state / feeling of remorse and wants to express it.

Essential condition – all parties recognize the utterance as a speech act

- E.g., speaker says this such that the listener can actually hear and decide whether to accept the apology or not.

The King of France is Bald (Again)

Is

The King of France is bald.

true or false?

New answer: it's not so important. This statement is **infelicitous**.

- **Presupposition failure!** This statement takes for granted that there is a King of France, but there is no King of France that can be bald.

Speech Act Theory

Austin (1962) developed his theory of speech acts and applied it to all statements, not just performative utterances. Main claims:

- When you “say” something, you are *also performing an action*.
- We can analyze this action at multiple levels:
 - **Locution, illocution, perlocution**

Locutionary Act

The speaker emitted sounds, typed out/formed words into sentences according to the grammar of a language, and that language has some literal meaning.

Can you pass me the salt?

Speaker asks if listener has the capability of moving the container of salt to the speaker.

Illocutionary Act

The speaker has an intended meaning that they want to convey to the listener. What's **in** the utterance.

Can you pass me the salt?

Polite request to the listener to pass the salt to the speaker.

Perlocutionary Act

By the speaker's utterance, a change in the world is effected.

Can you pass me the salt?

The container of salt is moved from someplace close to the listener, to some place close to the speaker.

A Bixi Ad



(Summer is never over.)

What does this mean?

Different Kinds of Meaning

Locutionary meaning (literal meaning)

- The season of summer extends forever into the future.

Illocutionary force (intended meaning)

- Some aspect related to summer (e.g., fun, outdoor activities) can go on beyond the summer season.
- Bixi bikes are now available for rent year-round.

Perlocutionary effect (effect on listener)

- You are more aware of Bixi, and might consider subscribing.

Rational Speech Act Theory

A computational approach that implements some aspects of pragmatic theories (Frank and Goodman, 2012)

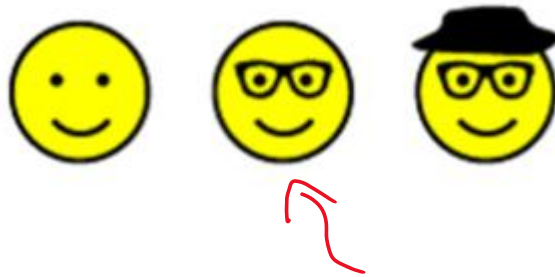
- Developed in cognitive science research

Key features:

- Model communicative context
- Agents with intents and beliefs
- Goal of communication is for speaker and listener to come to mutual understanding of communicative intent

RSA Basics and Assumptions

Language use as an instance of a **signalling game**

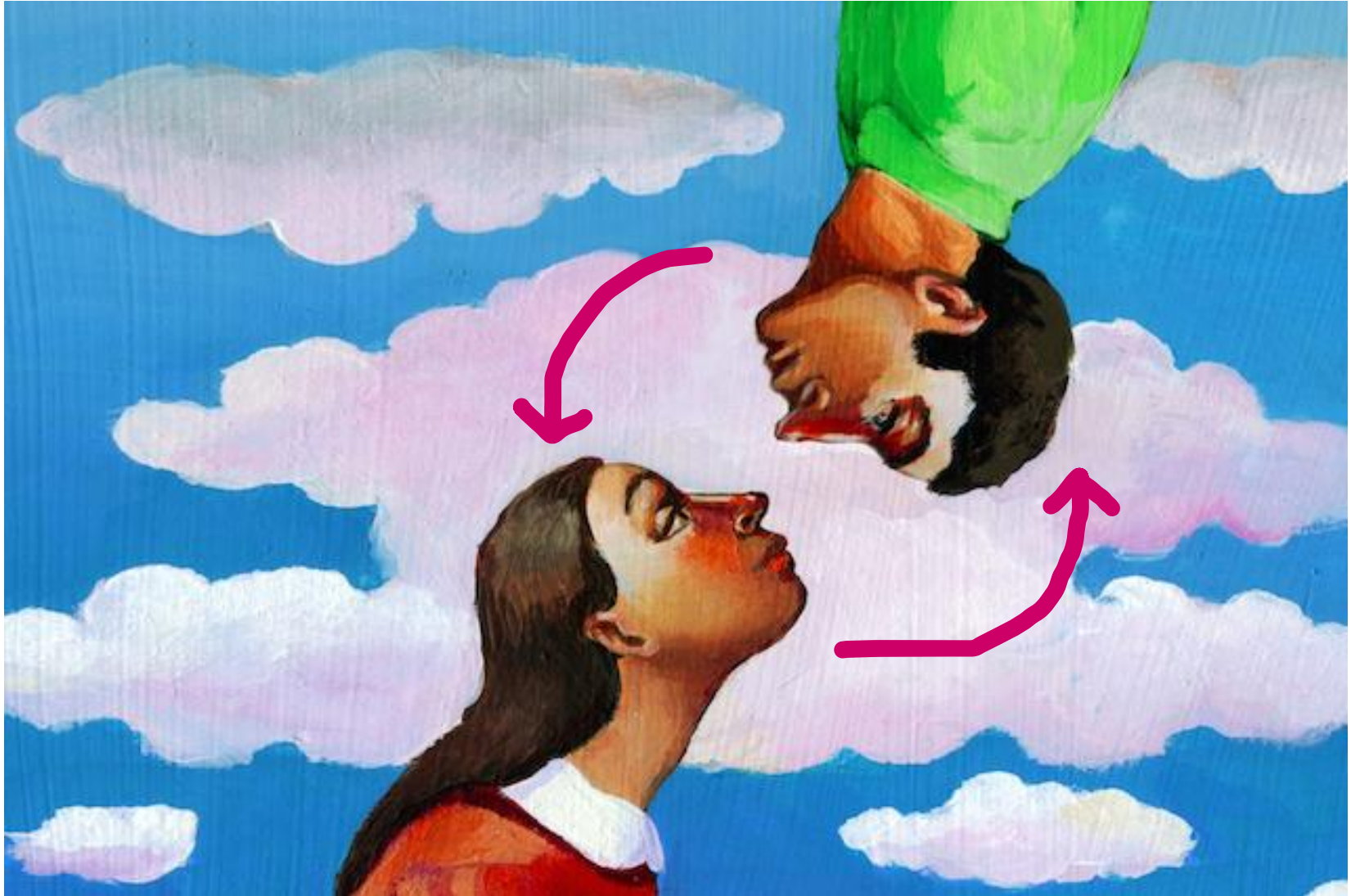


Speakers and listeners (**interlocutors**) reason about an explicitly defined set of utterances U and possible meanings, M .

- U : {glasses, hat, nothing}

- M : { , ,  }

Literal vs. Pragmatic Interlocutors



Literal vs. Pragmatic Interlocutors

Literal listener

- Interprets utterance of speaker literally
- *Hypothetical* – Serves as basis of recursive reasoning process

Pragmatic speaker

- Reasons about this literal listener
- Chooses an utterance from the set based on balancing informativeness and cost

Pragmatic listener

- Chooses most likely intended meaning based on prior subjective beliefs and their expectations about the pragmatic speaker's behaviour

Literal Listener

$$P_{L_0}(m|u) \stackrel{\text{prop}}{=} \delta_{m \in [[u]]} \cdot P(m).$$

Probability of meaning m given utterance u is proportional to the meaning's prior, as long as it is compatible with the utterance's literal meaning.

Prior Probability

How likely is the speaker to intend to say some meaning, without any context?



For our example, assume uniform prior.

$$P(m) = \frac{1}{3}$$

Worked Example

$$P_{L_0}(m|u) \stackrel{\text{prop}}{=} \delta_{m \in [[u]]} \cdot P(m).$$

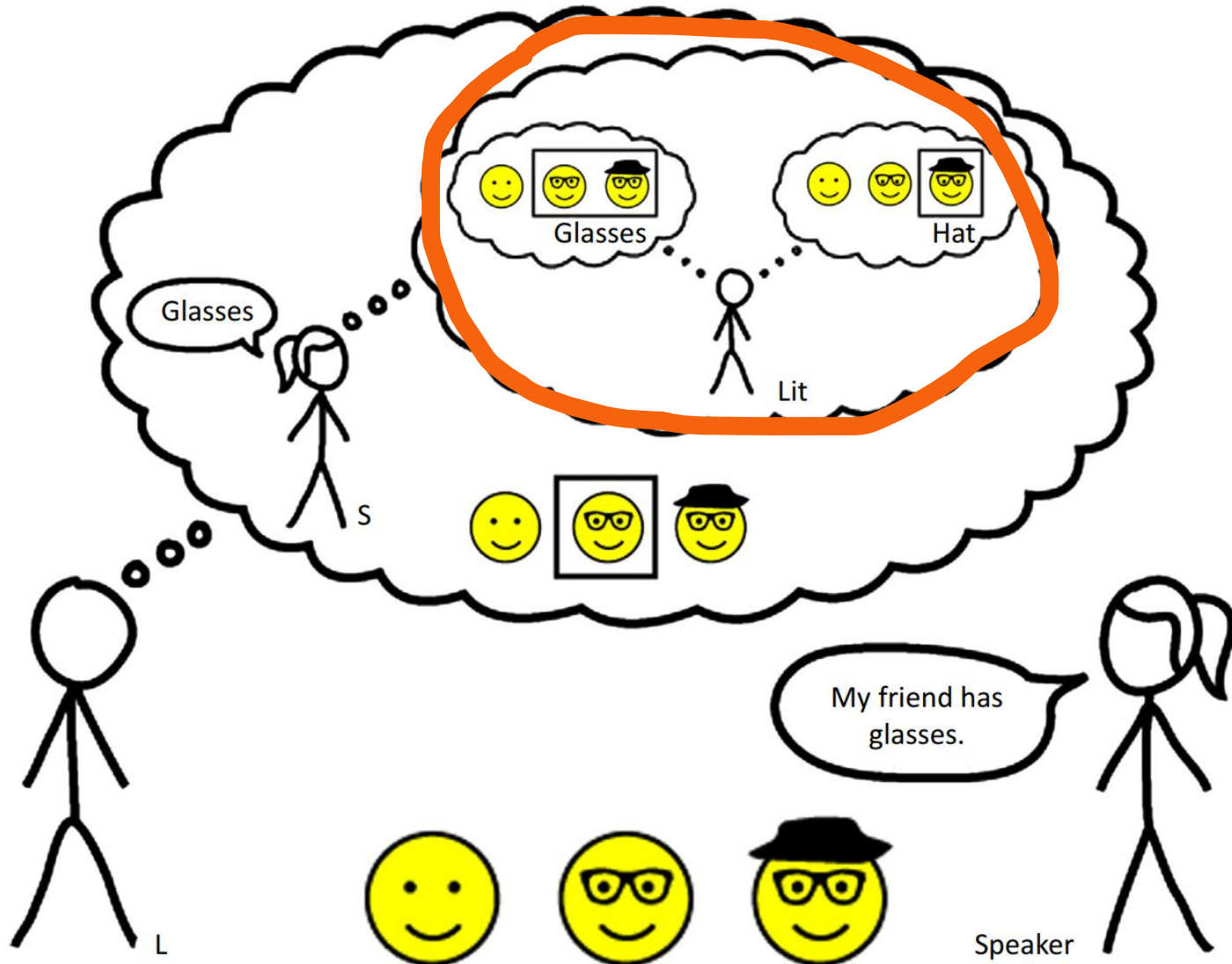


$$P_{L_0}(\cdot | \text{"glasses"})$$

$$P_{L_0}(\cdot | \text{"hat"})$$

$$P_{L_0}(\cdot | \text{"nothing"})$$

My friend has glasses



Pragmatic Speaker

Decide what to say based on reasoning about literal listener:

$$P_{S_1}(u|m) \propto \exp(\alpha \cdot U(u; m)).$$

Where $U(u, m)$ is a utility function defined by:

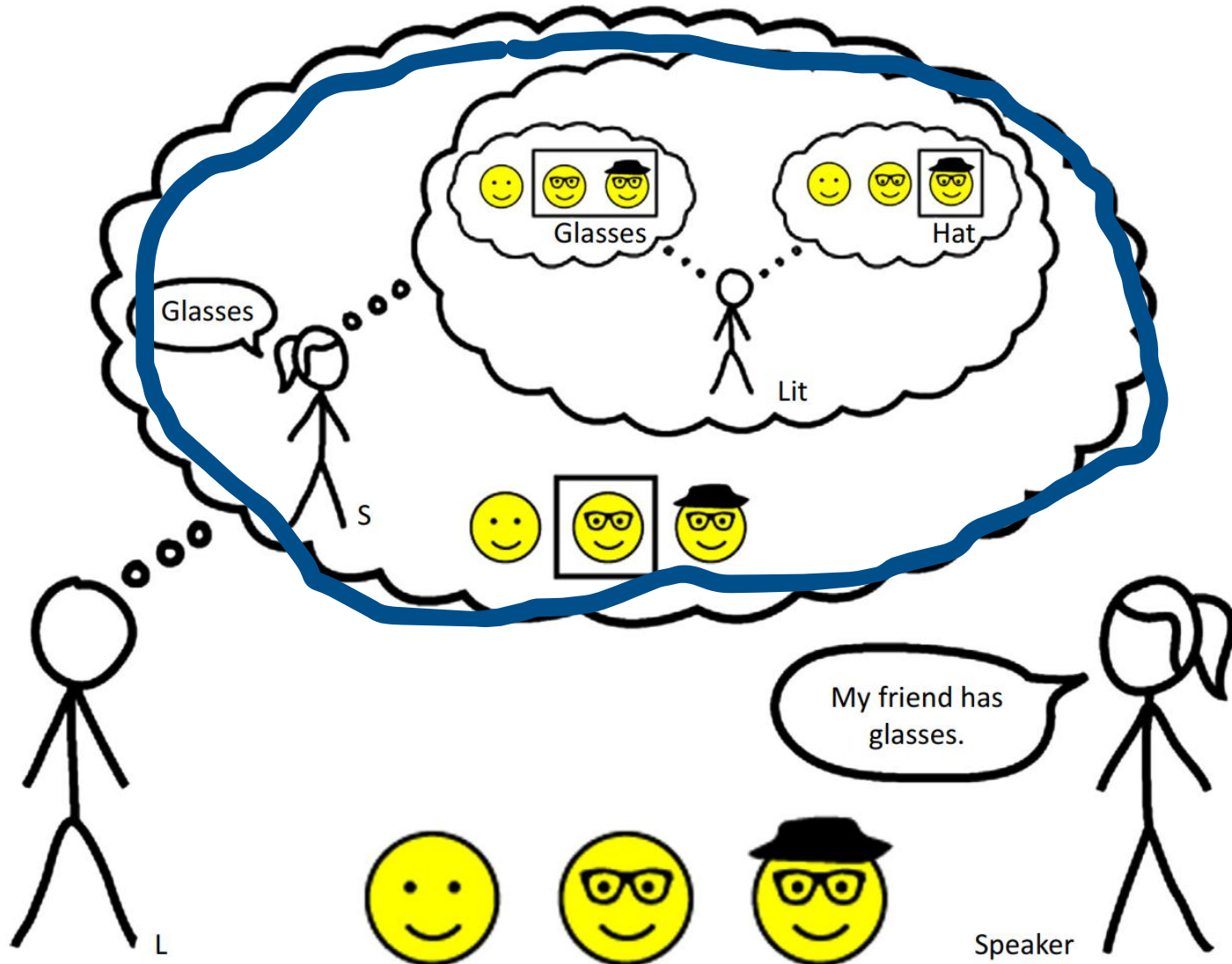
$$U(u, m) = \ln P_{L_0}(m|u) - \text{cost}(u).$$

Cost(u) is some cost function such as one based on length

Let's assume 0 cost for simplicity, alpha = 1.

$$P_{S_1}(u|m) \propto P_{L_0}(m|u)$$

Pragmatic Speaker Example



Worked Example (Cont'd)

$$P_{S_1}(u|m) \propto P_{L_0}(m|u)$$

$$P_{S_1}(\text{"glasses"} | \quad)$$



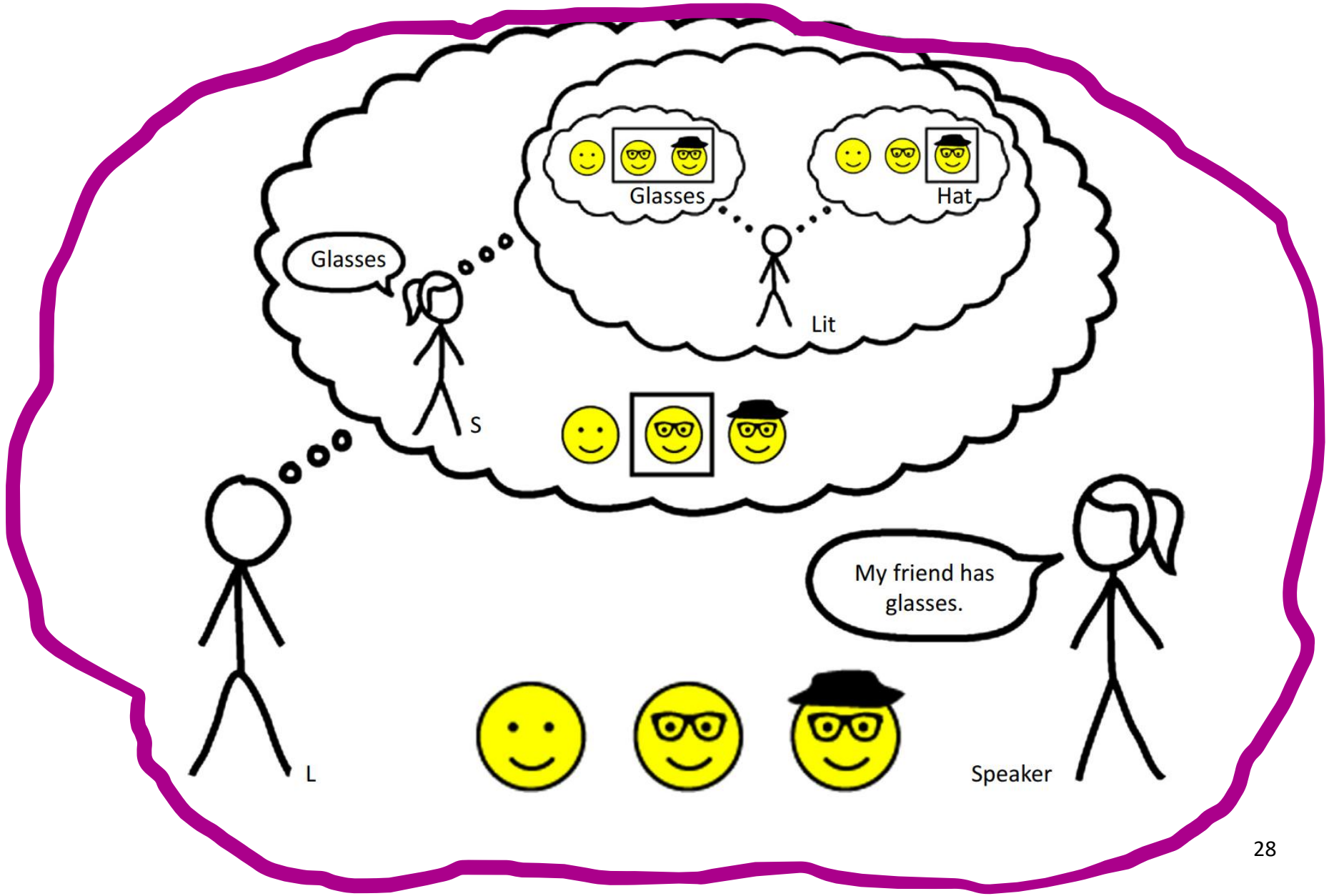
Pragmatic Listener

Combines evidence from reasoning about pragmatic speaker with prior beliefs:

$$P_{L_1}(m|u) \propto P_{S_1}(u|m) \cdot P(m).$$

This is an application of Baye's Rule!

Pragmatic Listener Example



Worked Example (Cont'd)

$$P_{L_1}(m|u) \propto P_{S_1}(u|m) \cdot P(m).$$



$$P_{L_1}(\cdot | \text{"glasses"})$$

A Second Example: Scalar Implicature

“Alex ate some of the cookies.”



There were four cookies to start with. How many cookies do you think Alex ate?

A Second Example: Scalar Implicature

Alex ate (none / some / all) of the cookies.



Let's set up the problem and work through a round of RSA.

Definitions and Prior

m: number of cookies eaten (0 – 4)

- Uniform prior: $P(m) = 0.2$

u:

- “none”: 0 cookies eaten
- “some”: 1 – 4 cookies eaten
- “all”: 4 cookies eaten

Literal Listener

$$P_{L_0}(m|u) \stackrel{\text{prop}}{=} \delta_{m \in [[u]]} \cdot P(m).$$

Pragmatic Speaker

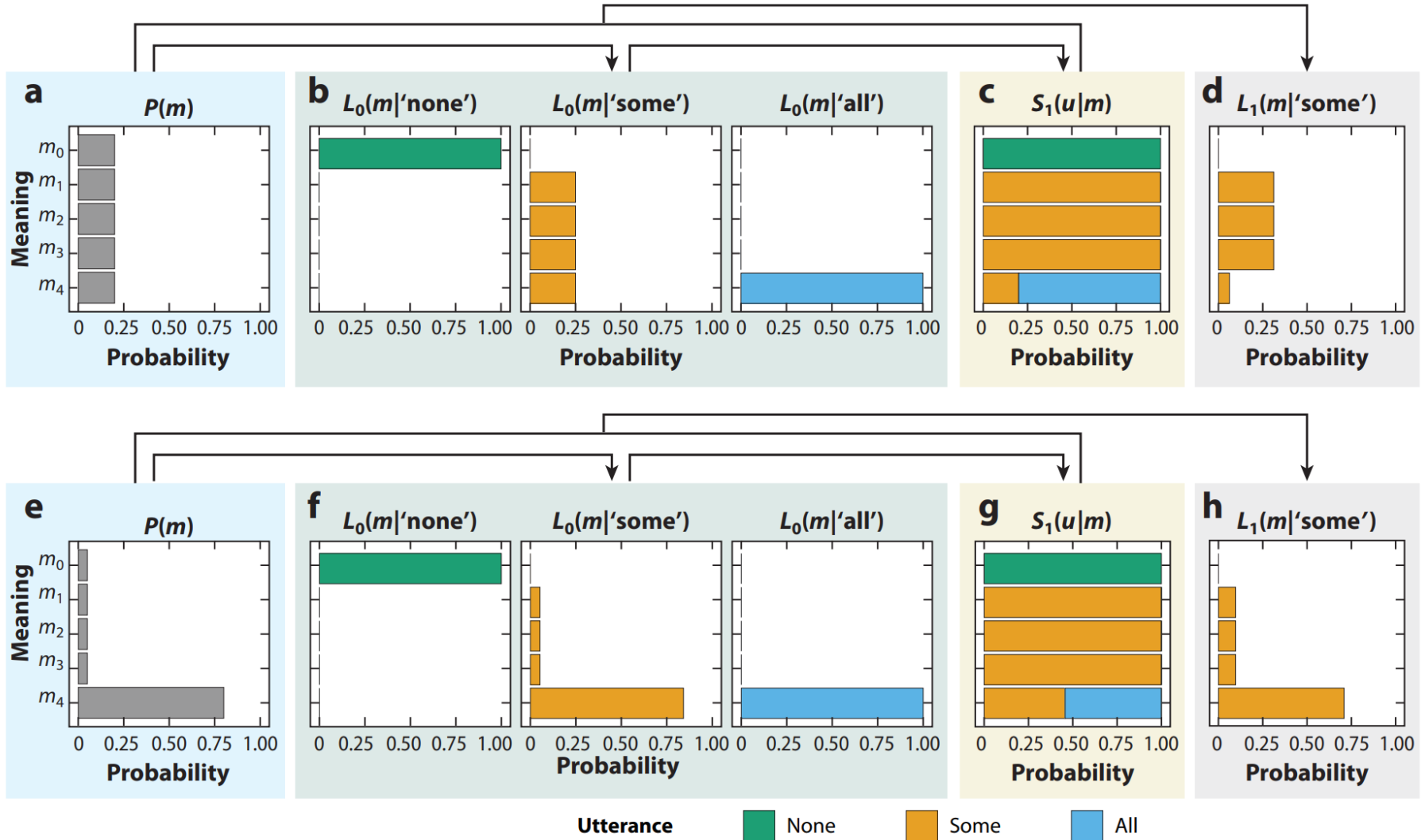
$$P_{S_1}(u|m) \propto \exp(\alpha \cdot U(u; m)).$$

$$U(u, m) = \ln P_{L_0}(m|u) - \text{cost}(u).$$

Pragmatic Listener

$$P_{L_1}(m|u) \propto P_{S_1}(u|m) \cdot P(m).$$

Answers, Also with Skewed Prior



Recap

Pragmatics is the study of intended meaning in context
Speech act theory gives us three levels of “meaning” or “effect”:

- Locution
- Illocution
- Perlocution

Rational speech act theory is a Bayesian theory of probabilistic pragmatics

- Models speakers and listeners reasoning about each others’ beliefs and intentions

Further Readings

- Goodman, Noah D., and Michael C. Frank. "Pragmatic language interpretation as probabilistic inference." *Trends in cognitive sciences* 20.11 (2016): 818-829.
- Austin, John Langshaw. *How to do things with words*. Harvard university press, 1975. (First published 1962, based on lectures delivered in 1955)
- Degen, Judith. "The rational speech act framework." *Annual Review of Linguistics* 9.1 (2023): 519-540.